

Subsystem Specification

Propulsion

Turn helm commands into bidirectional thrust on two clip-on pods, quietly, safely and without being defeated by weed.

Document	BOATY-SSS-PRP
Issue	Issue B (for review)
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Status	For review by the project owner
Parents	SRS Issue F, ADD Issue F, ICD Issue E, KCL Issue B
Content	7 allocated SRS requirements → 16 subsystem requirements (10 M, 6 S, 0 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	29 September 2026	ESC chosen: AM32 20 A, DShot 3D set by the helm (PRP-D06); ESC configuration PRP-D16; cost from the Key Component List.	Claude, owner decision (CR-04)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

Contents

1. Scope	3
1.1 Purpose	3
1.2 Boundary	3
1.3 Breakdown	3
2. Applicable documents	3
3. Interfaces	3
4. Parent requirements	4
5. Subsystem requirements	5
5.1 Performance	5
5.2 Control interface and failure behaviour	5
5.3 Guarding and weed	5
5.4 Mechanical and handling	5
6. Design analysis: thrust and power	6
7. Design constraints and budgets	6
8. Verification and open items	6

1. Scope

1.1 Purpose

Turn helm commands into bidirectional thrust on two clip-on pods, quietly, safely and without being defeated by weed.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Two thruster pods (motor, prop, shroud/guard, strut, dovetail, latch) • Two bidirectional ESCs (in the box) • Pod connectors and motor leads	• Motor power (PWR) • Motor commands and speed control (HLM) • Transom mount features (HUL)

1.3 Breakdown

Item	Description
PRP-1 Thruster pod ×2	2204-2208 outrunner running submerged; printed prop in a full shroud
PRP-2 ESC ×2	20-30 A, BLHeli_S/AM32, bidirectional, DShot
PRP-3 Pod connector ×2	Keyed 3-pole IP68

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue D	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue C	Interface definitions (all listed in section 3 apply in full)
BOATY-FMEA-001 Issue B	Failure modes and actions carried into this issue

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-05 Motor commands	Helm	Conforms	\$6
IF-06 Motor power	Power	Conforms	\$6
IF-17 Pod mount	Hull & structure	Owner	\$8

4. Parent requirements

Primary (7): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
NAV-001	M	The boat shall steer by differential thrust from two thrusters, and be able to go forward and in reverse.	PRP-D01, PRP-D02, PRP-D06, PRP-D15
FS-008	M	Helm failure: if motor commands stop arriving from the helm, the motors shall stop within 1 s.	PRP-D07, PRP-D08, PRP-D16
MEC-008	S	Swapping a thruster pod shall take under 2 min. Pod connectors shall be waterproof and keyed.	PRP-D12
MEC-010	M	Propellers shall be guarded so that an 8 mm diameter probe cannot touch a blade.	PRP-D09
MEC-011	S	Thruster pods shall include weed-shedding features: a swept leading edge and a guard shape that sheds weed.	PRP-D10
ENV-002	M	The boat shall make at least 0.5 m/s headway into a 5.4 m/s wind (top of Beaufort 3).	PRP-D02
ENV-005	S	The boat shall keep operating in sparse surface weed, recovering through FS-005 and FS-006.	PRP-D10, PRP-D11

Supporting (8): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

NAV-002, NAV-003, FS-006, PWR-009, CHD-001, CHD-002, CHD-003, CHD-004

5. Subsystem requirements

Requirements are numbered PRP-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Performance

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D01	Two identical pods, port and starboard, shall each give forward and astern thrust on command.	M	D	POOL	NAV-001
PRP-D02	Combined static forward thrust shall be ≥ 4 N, and astern ≥ 1.5 N. Sized from the section 6 estimate: ≈ 1.1 N needed for 0.5 m/s into 5.4 m/s wind, with a margin of 3.5.	M	T	BENCH	ENV-002, NAV-001
PRP-D03	The boat shall reach 1.0 m/s in calm water at $\leq 50\%$ throttle.	M	T	POOL	NAV-002
PRP-D04	Combined electrical input at 1.0 m/s cruise shall be ≤ 10 W.	S	T	POOL	PWR-009
PRP-D05	Continuous current shall be ≤ 8 A per ESC at full throttle, measured.	M	T	BENCH	IF-06

5.2 Control interface and failure behaviour

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D06	The ESCs (AM32 20 A, KCL KC-03) shall accept commands per IF-05 (DShot300, 3D mode; PWM fallback).	M	T	BENCH	IF-05, NAV-001
PRP-D07	Each ESC shall stop its motor ≤ 1 s after command pulses stop.	M	T	BENCH	FS-008, V-09
PRP-D08	After signal loss, an ESC shall not restart until it sees a neutral command.	M	T	BENCH	FS-008

5.3 Guarding and weed

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D09	Each prop shall be fully shrouded, so that an 8 mm probe cannot touch a blade from any direction.	M	T	BENCH	MEC-010
PRP-D10	Guard and strut leading edges shall be swept back $\geq 30^\circ$, with no forward-facing hooks. Inlet bars shall run at $\geq 45^\circ$ to the flow. Prop blades shall have swept leading edges.	S	I	BENCH	MEC-011, ENV-005
PRP-D11	In sparse surface weed, a pod fouled at the inlet shall be clearable by ≤ 3 reverse bursts (FS-006) in ≥ 2 of 3 trials.	S	T	LAKE	ENV-005

5.4 Mechanical and handling

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D12	Pods shall mount per IF-17 and be swappable in < 2 min, tool-free, with keyed waterproof connectors.	S	T	BENCH	MEC-008, IF-17
PRP-D13	Each pod shall weigh ≤ 75 g including the motor.	S	T	BENCH	MEC-002
PRP-D14	Pod shells and latches are child-handled parts (HUL-D22/D23 apply). The prop shall not be reachable while the pod is fitted or unfitted.	M	I	BENCH	CHD-001, CHD-002, CHD-003
PRP-D16	ESC configuration: 3D mode set by the helm at boot (SERVO_BLH_3DMASK); low-voltage cut-off disabled, because battery protection belongs to the helm and BMS; BEC output left unconnected. AM32 stops the motor 0.5 s after signal loss and needs > 1 s of zero command to re-arm (firmware source, confirmed on the bench).	M	I	BENCH	IF-05, FS-008, V-09
PRP-D15	Motors are consumables: rinse and dry after each session. Expected life ≥ 10 h submerged before replacement.	S	A	LAKE	NAV-001

6. Design analysis: thrust and power

Case	Working	Result
Water drag at 0.5 m/s	$\frac{1}{2}\rho v^2 \cdot S \cdot C_t$, $S = 0.18 \text{ m}^2$, $C_t = 0.012$, $\times 1.5$ form	0.41 N
Wind drag, 5.4 m/s head wind	$\frac{1}{2}\rho_{\text{air}}(5.9 \text{ m/s})^2 \cdot 0.035 \text{ m}^2 \cdot 1.1$	0.82 N
Needed for ENV-002	sum	$\approx 1.2 \text{ N}$
Specified thrust (PRP-D02)	$\geq 4 \text{ N}$ combined	margin $\approx 3.3\times$
Drag at 1.0 m/s cruise	with wave-making $\times 1.3$	$\approx 1.4 \text{ N} \rightarrow 1.4 \text{ W}$ at the water
Electrical at cruise	overall efficiency $\approx 15\%$	$\approx 9 \text{ W}$ (PRP-D04 $\leq 10 \text{ W}$)
Small printed props in shrouds are inefficient. The 15% figure is deliberately pessimistic and gets replaced by bench thrust-stand data.		

7. Design constraints and budgets

- ADD T3/T5: submerged brushless, differential thrust, no rudder
- Airboat pod is a future swap-in module (same IF-17)

Budget	Allocation
Mass allocation	$\leq 200 \text{ g}$ (estimate 180 g)
Cost allocation	£34 (motors £12, 2 $\times$ AM32 ESC £22)
Power	$\approx 8 \text{ W}$ at cruise (estimate); $\leq 16 \text{ A}$ total continuous

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	8	-	3	-	-	11
POOL	2	1	-	-	-	3
LAKE	1	-	-	1	-	2

Ref	Open item
TBC-05	ESC DShot and 3D-mode support (IF-05)
TBC-06	Measured motor current (IF-06)
TBC-12	Pod connector part number (IF-17)
V-09	ESC stop on signal loss $\leq 1 \text{ s}$